

Detailed study 2 – Photonics

23	band gap energy	$E = \frac{hc}{\lambda}$
24	Snell's law	$n_1 \sin \theta_1 = n_2 \sin \theta_2$

Detailed study 3 – Sound

25	speed, frequency and wavelength	$v = f\lambda$
26	intensity and levels	sound intensity level $(\text{in dB}) = 10 \log_{10} \left(\frac{I}{I_0} \right)$ where $I_0 = 1.0 \times 10^{-12} \text{ W m}^{-2}$

Prefixes/Units

$$\text{p} = \text{pico} = 10^{-12}$$

$$\text{n} = \text{nano} = 10^{-9}$$

$$\mu = \text{micro} = 10^{-6}$$

$$\text{m} = \text{milli} = 10^{-3}$$

$$\text{k} = \text{kilo} = 10^3$$

$$\text{M} = \text{mega} = 10^6$$

$$\text{G} = \text{giga} = 10^9$$

$$\text{t} = \text{tonne} = 10^3 \text{ kg}$$